

PROBLEM STATEMENT

Power & Trigonometric Chain Rule

Find the derivative with respect to x of the function:

$$y = \sin^3(2x + 3)$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক ঘনক ফাংশনটিকে বন্ধনী আকারে লিখি (Expressing in Explicit Power Form)

$$y = [\sin(2x + 3)]^3$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} ([\sin(2x + 3)]^3)$$

STEP 3 বহিঃস্থ ঘাত ৩-এর জন্য শক্তি বিধি (Power Rule) প্রয়োগ করি (Applying Power Rule to Outer Exponent)

$$\frac{dy}{dx} = 3[\sin(2x + 3)]^{3-1} \cdot \frac{d}{dx} [\sin(2x + 3)] = 3 \sin^2(2x + 3) \cdot \frac{d}{dx} [\sin(2x + 3)]$$

STEP 4 সাইন ফাংশন ও রৈখিক কোণ $(2x + 3)$ -এর অন্তরজ নির্ণয় করি (Differentiating Sine & Linear Angle via Chain Rule)

$$\frac{d}{dx} [\sin(2x + 3)] = \cos(2x + 3) \cdot \frac{d}{dx} (2x + 3) = \cos(2x + 3) \cdot (2) = 2 \cos(2x + 3)$$

STEP 5 ধ্রুবক গুণ করে চূড়ান্ত অন্তরজের সরল রূপ পাই (Combining Constant Factors into Final Derivative)

$$\frac{dy}{dx} = 3 \sin^2(2x + 3) \cdot [2 \cos(2x + 3)] = 6 \sin^2(2x + 3) \cos(2x + 3)$$

FINAL ANSWER $\frac{dy}{dx} = 6 \sin^2(2x + 3) \cos(2x + 3)$ (or, $3 \sin(2x + 3) \sin(4x + 6)$)

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Three-Stage Chain Rule Composition:** Let $u = \sin(2x + 3)$ and $v = 2x + 3$. Then $y = u^3$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = (3u^2) \cdot (\cos v) \cdot (2) = 6 \sin^2(2x + 3) \cos(2x + 3)$$
- Trigonometric Power Rule:** $\frac{d}{dx} [\sin^n(ax + b)] = n \cdot a \cdot \sin^{n-1}(ax + b) \cos(ax + b)$.
- Double-Angle Equivalent:** $2 \sin \theta \cos \theta = \sin(2\theta) \implies 6 \sin^2 \theta \cos \theta = 3 \sin \theta \sin(2\theta)$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Missing Linear Factor:** Forgetting to multiply by $\frac{d}{dx}(2x + 3) = 2$, mistakenly leaving the answer as $3 \sin^2(2x + 3) \cos(2x + 3)$.
- Power Notation Misplacement:** $\sin^3(2x + 3) = [\sin(2x + 3)]^3 \neq \sin((2x + 3)^3)$.
- Differentiating Angle as Constant:** Confusing $\frac{d}{dx}(2x + 3) = 2$ with 0 by accidentally dropping x .

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Real Domain:** Trigonometric sine and linear polynomials are defined on the whole real axis:
$$\text{Domain} = (-\infty, \infty) = \mathbb{R}$$
- Bounded Range:** Since $\sin(2x + 3) \in [-1, 1]$, $y = \sin^3(2x + 3) \in [-1, 1]$.
- Differentiability:** Entirely smooth and infinitely differentiable (C^∞) everywhere on $\mathbb{R}$.

BOX 4: ALTERNATIVE METHOD VIA TRIPLE-ANGLE IDENTITY

- Using identity $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta \implies \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$:
$$y = \frac{3}{4} \sin(2x + 3) - \frac{1}{4} \sin(6x + 9)$$
- Differentiating term-by-term:
$$\frac{dy}{dx} = \frac{3}{4} \cos(2x + 3) \cdot (2) - \frac{1}{4} \cos(6x + 9) \cdot (6) = \frac{3}{2} \cos(2x + 3) - \frac{3}{2} \cos(6x + 9)$$
- Using sum-to-product formulas, this identically simplifies back to $6 \sin^2(2x + 3) \cos(2x + 3)$.

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (Power Representation & Chain Strategy):** 1.0 Mark
- Step 3 (Power Rule Differentiation):** 1.0 Mark
- Step 4 (Trig Derivative with Linear Multiplier):** 1.0 Mark
- Step 5 & Final Answer:** 1.0 Mark
- Total Allocation:** 4.0 Marks